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(54) **FASCIA DOOR AND WASHER FLUID RESERVOIR**

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B60R 19/48 (2006.01)

(52) **U.S. Cl.**
CPC ***B60S 1/50*** (2013.01); *B60R 19/48* (2013.01)

(58) **Field of Classification Search**
CPC B60S 1/50; B60R 19/023; B60R 19/18; B60R 2019/1886; B60R 13/04; B60R 19/48; B60K 2015/0523
USPC 296/1.07, 193.08, 1.08; 293/102, 120, 293/121
See application file for complete search history.

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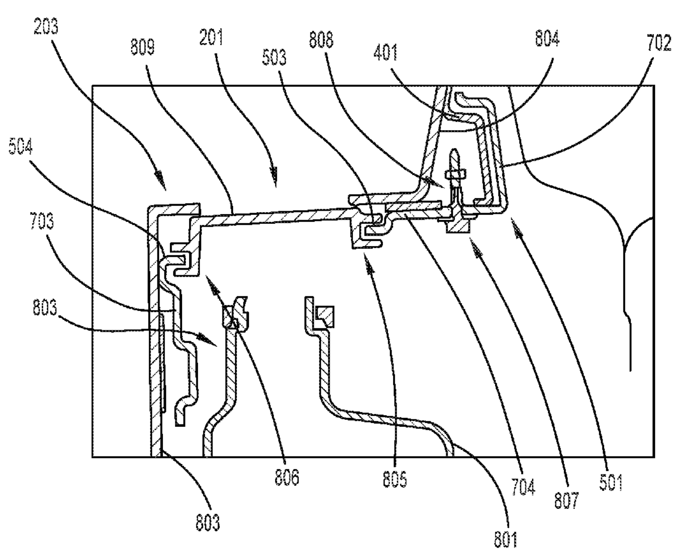
(57) **ABSTRACT**

A fascia door assembly including: a fascia door support bracket having a support bracket opening; a fascia door bracket coupled to the fascia door support bracket and having a bracket opening; and a fascia door engaged with the fascia door bracket in the bracket opening, the fascia door having an opened position in the bracket opening and a closed position in the bracket opening.

18 Claims, 6 Drawing Sheets



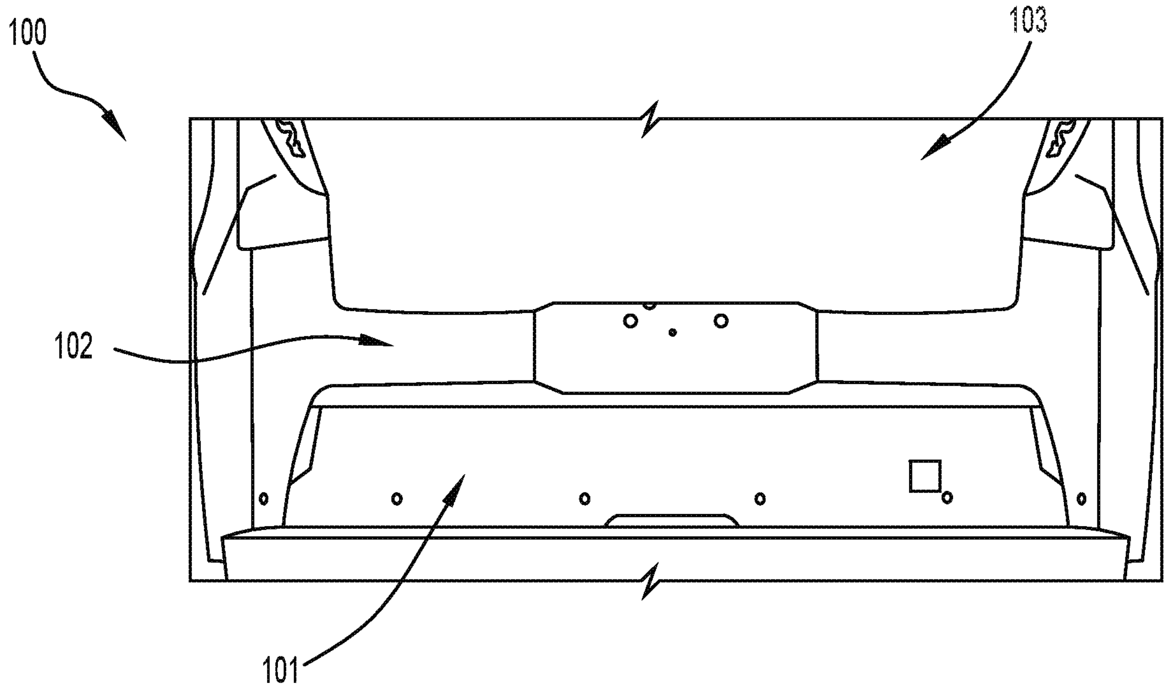


FIG. 1

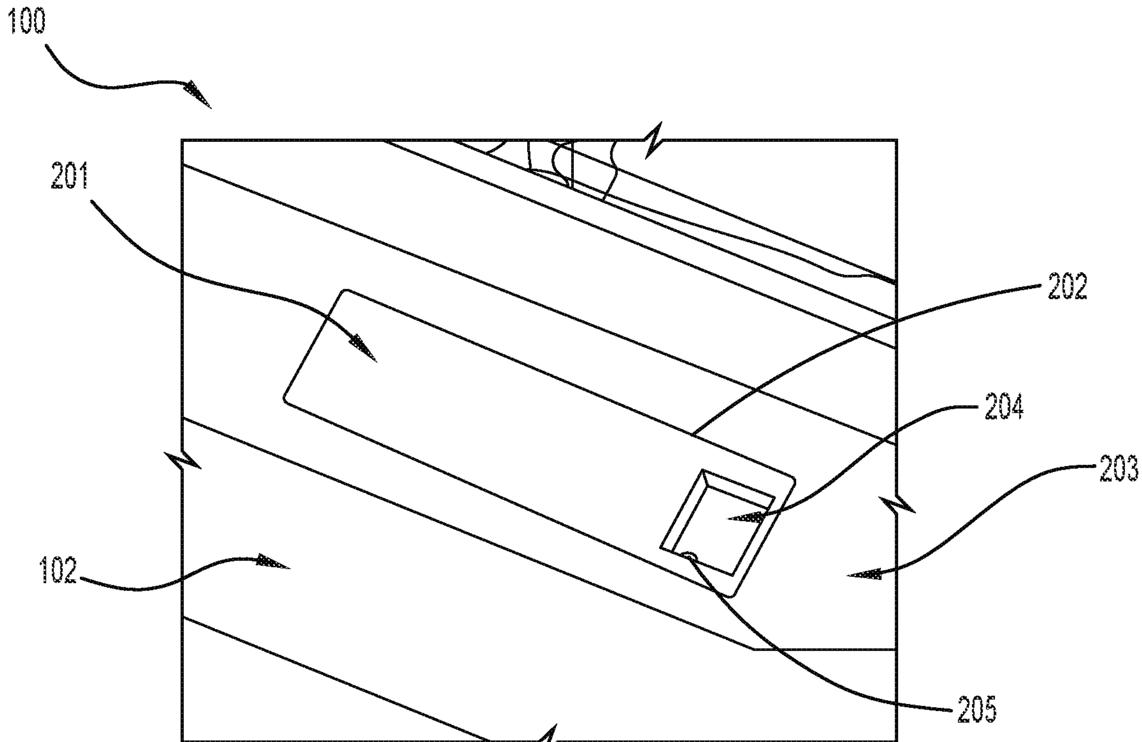


FIG. 2

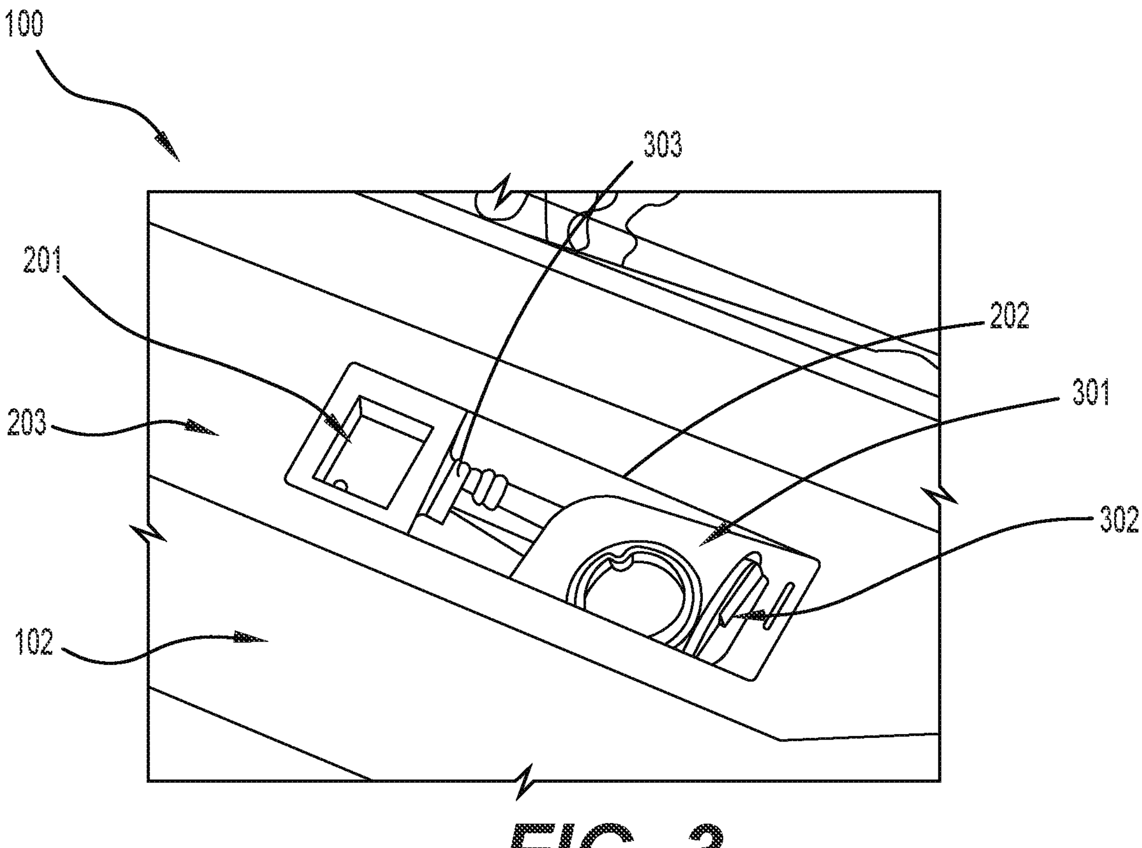


FIG. 3

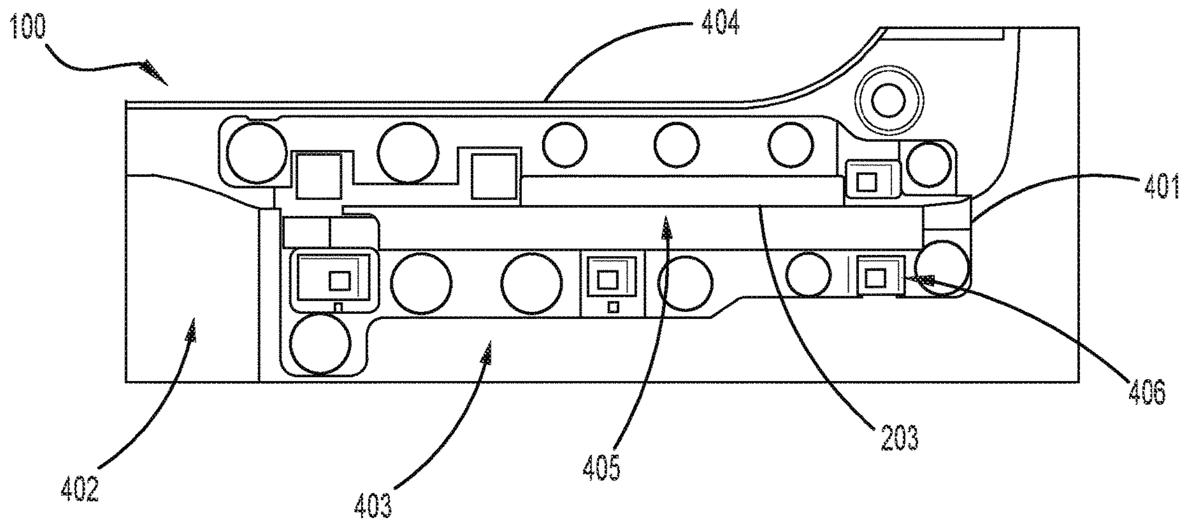


FIG. 4

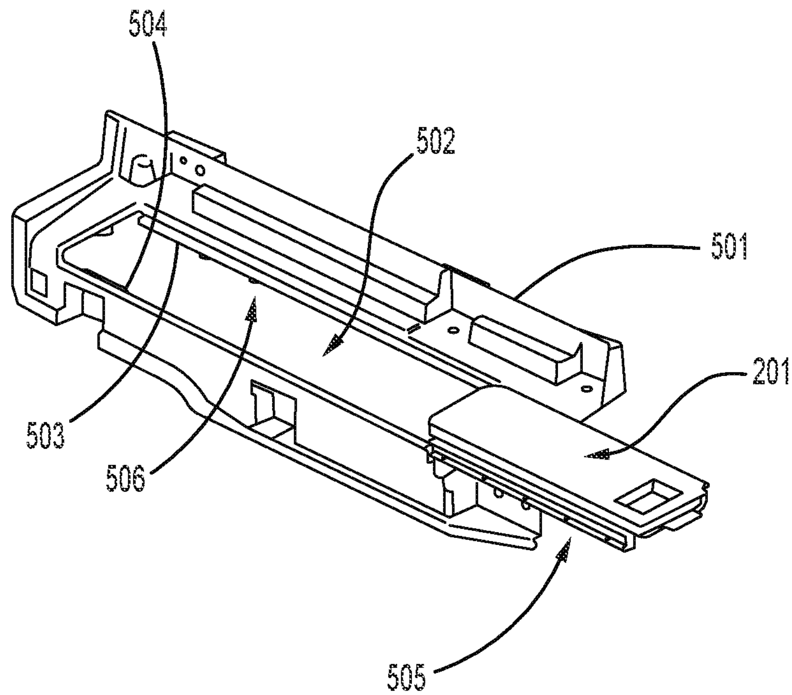


FIG. 5

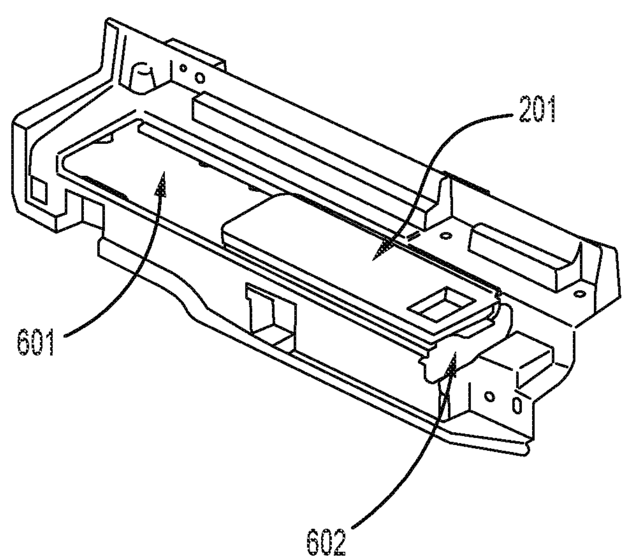


FIG. 6

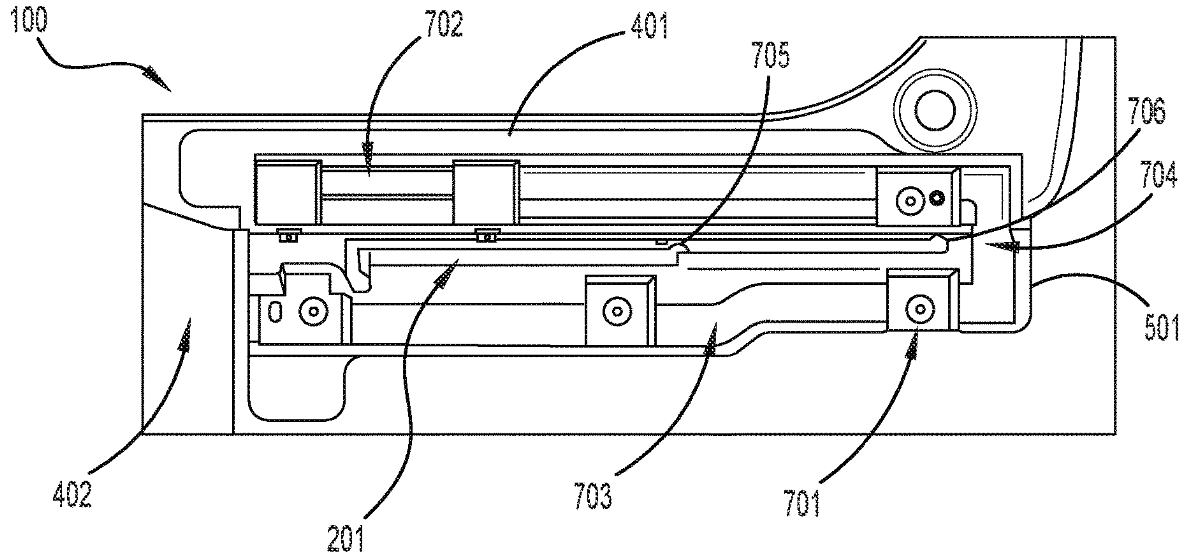

100
702
401
705
706
704
501
402
201
703
701

FIG. 7

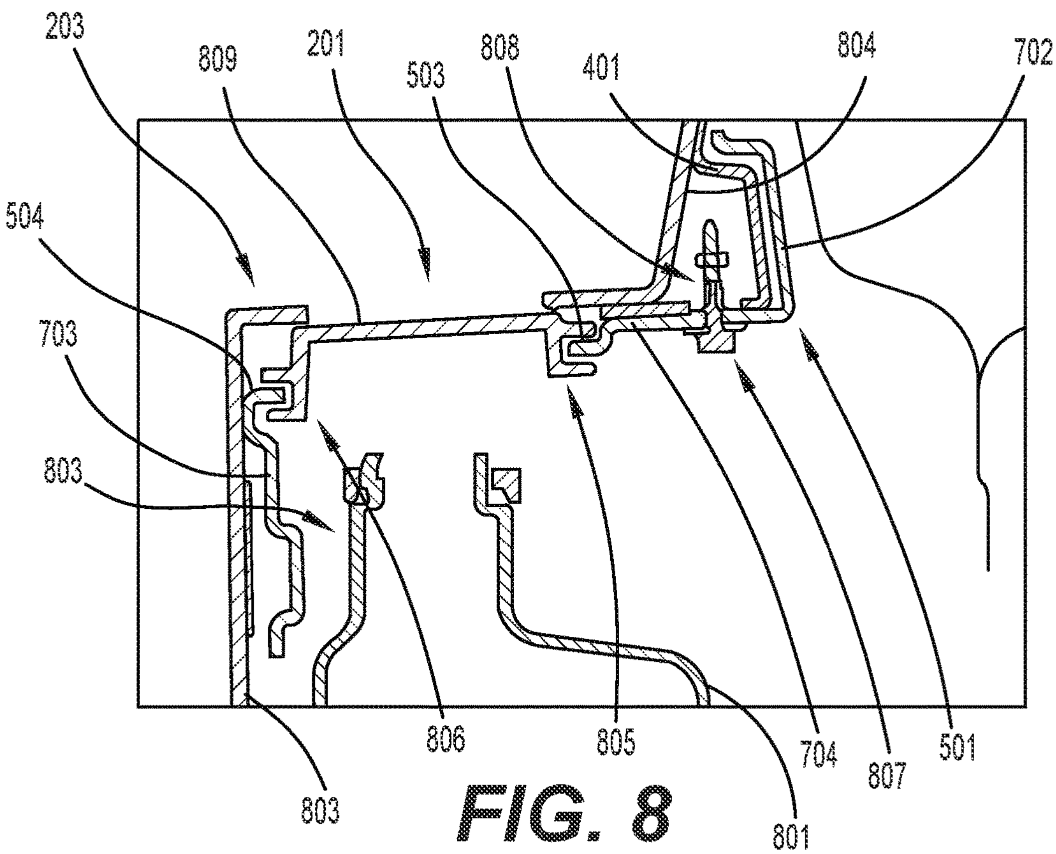

203
809
201
503
808
401
804
702
504
703
803
806
805
704
807
501
803
801

FIG. 8

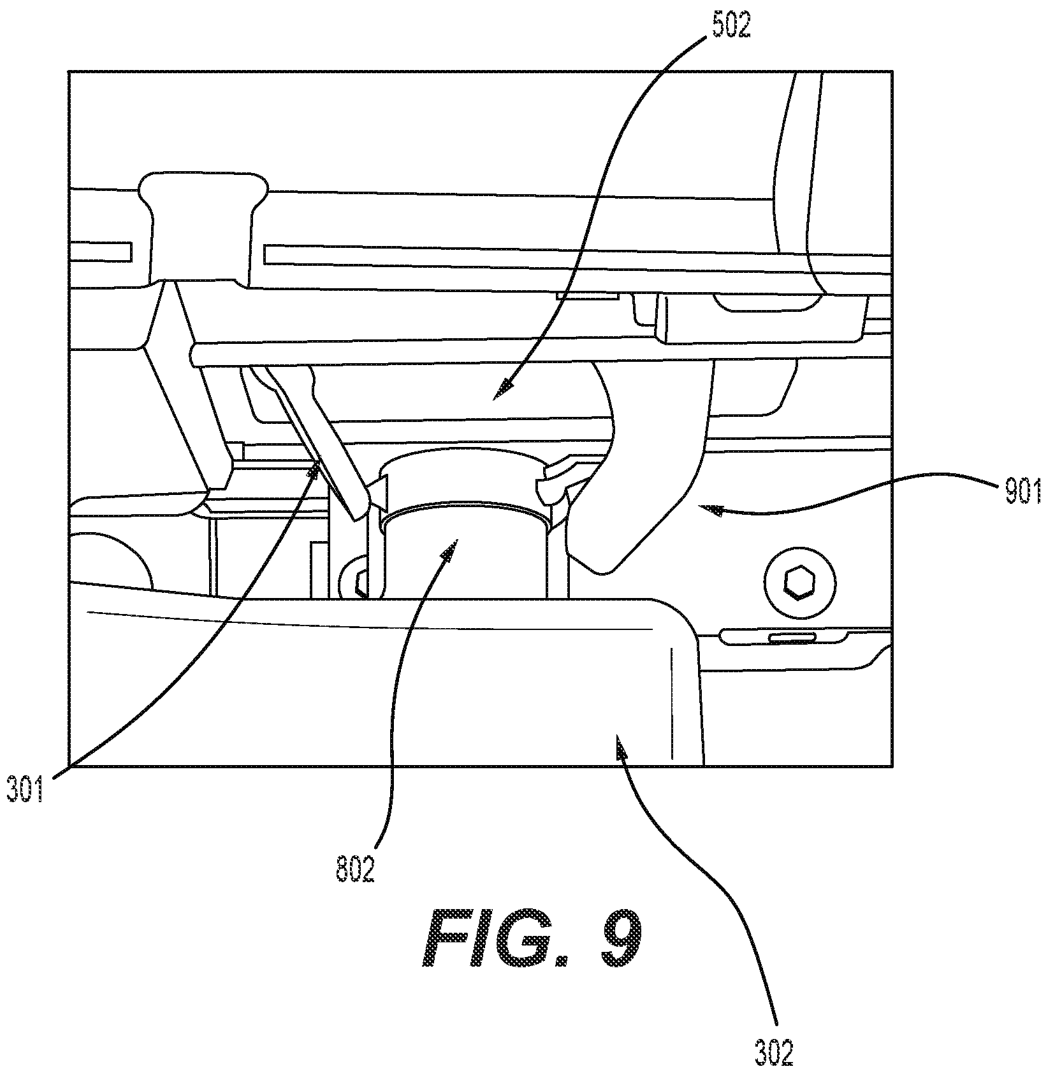


FIG. 9

FASCIA DOOR AND WASHER FLUID RESERVOIR

BACKGROUND

1. Field of the Disclosure

The present disclosure relates to a fascia door concealing a washer fluid reservoir, and more particularly to a fascia door concealing a washer fluid reservoir mounted behind a rear fascia of a vehicle.

2. Description of Related Art

In many cars, a windshield washer fluid tank is located near a front wheel well, e.g., behind a headlight or in front of a bulkhead or firewall.

The windshield washer fluid tank needs to have convenient accessibility for refill, serviceability, etc. Conventional positioning below a hood gives the advantage of accessibility and safety.

As car design and construction advances, with the increase in demand for electric vehicles, positioning below the hood is no longer convenient, as the space is increasingly used for storage or for high powered electronics.

There is a need in the art for a system and method that positions the windshield washer fluid reservoir in accessible and safe locations for modern vehicles.

SUMMARY OF THE INVENTION

For vehicles with upright windshields, washer fluid usage is relatively higher than for a typical vehicle with an angled windshield. Other types of vehicles may have similarly higher washer fluid usage, such as to clean sensors for parking or navigation, or to clean rear windows. In each of these vehicles, frequent access to a washer fluid reservoir may be needed.

In one aspect, the disclosure provides a fascia door assembly including: a fascia door support bracket having a support bracket opening; a fascia door bracket coupled to the fascia door support bracket and having a bracket opening; and a fascia door engaged with the fascia door bracket in the bracket opening, the fascia door having an opened position in the bracket opening and a closed position in the bracket opening.

In one or more aspects, the disclosure provides a vehicle comprising: a rear fascia having an opening in a horizontal portion of the rear fascia; a washer fluid reservoir behind the rear fascia and below the opening; and a fascia door, wherein the washer fluid reservoir is concealed below the fascia door in a closed position and revealed by movement of the fascia door to an open position.

In another aspect, the disclosure provides a vehicle comprising: a rear fascia having an opening; a washer fluid reservoir behind the rear fascia; and a fascia door assembly comprising: a fascia door support bracket attached to an interior surface of the rear fascia and having a support bracket opening; a fascia door bracket coupled to the fascia door support bracket and having a bracket opening coincident with the opening in the rear fascia and the support bracket opening; and a fascia door engaged with the fascia door bracket in the bracket opening, wherein the washer fluid reservoir is concealed below the fascia door in the opening in the rear fascia.

Other systems, methods, features and advantages of the disclosure will be, or will become, apparent to one of ordinary skill in the art upon examination of the following figures and detailed description. It is intended that all such additional systems, methods, features and advantages be included within this description and this summary, be within the scope of the disclosure, and be protected by the following claims.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure can be better understood with reference to the following drawings and description. The components in the figures are not necessarily to scale, emphasis instead being placed upon illustrating the principles of the disclosure. Moreover, in the figures, like reference numerals designate corresponding parts throughout the different views.

FIG. **1** illustrates an embodiment of a vehicle including a rear fascia;

FIG. **2** illustrates an embodiment of a vehicle including a rear fascia including a fascia door in a closed position;

FIG. **3** illustrates an embodiment of a vehicle including a rear fascia including a fascia door in an open position;

FIG. **4** illustrates an embodiment of a fascia door support bracket attached to an interior surface of a rear fascia

FIG. **5** illustrates an embodiment of a fascia door engaged with a fascia door bracket **501**;

FIG. **6** illustrates an embodiment of a fascia door engaged with a fascia door bracket;

FIG. **7** illustrates an embodiment of a fascia door bracket engaged with a fascia door support bracket;

FIG. **8** illustrates an embodiment of cross-section of a fascia door assembly; and

FIG. **9** illustrates an embodiment of a fascia door and a washer fluid reservoir filler door.

DETAILED DESCRIPTION

According to some aspects, a fascia door conceals a washer fluid reservoir. According to one aspect, the washer fluid reservoir is positioned rear mounted, behind a rear fascia (e.g., a bumper cover) of a vehicle, for example to a chassis cross member, a bumper reinforcement bar, an impact bar, etc. According to one or more aspects, a filler door of the washer fluid reservoir is concealed by a fascia door.

FIG. **1** illustrates an embodiment of a vehicle including a rear fascia **100**. The rear fascia **100** may include a lower fascia portion **101**, an upper fascia portion **102**, and a horizontal portion (not shown) extending forward from the upper fascia portion **102**. The vehicle may include a lift gate **103**, which in a closed position conceals at least a portion of the horizontal portion.

FIG. **2** illustrates an embodiment of the vehicle including the rear fascia **100** including a fascia door **201** in a closed position and disposed in an opening **202** in the horizontal portion **203**. The fascia door **201** may be positioned to occupy an opening in the rear fascia **100**.

According to some aspects, the opening **202** of the rear fascia **100** is in a horizontal portion of the rear fascia **100** and access to the opening **202** depends on a position of a lift gate of the vehicle. For example, the opening **202** may be complete concealed by the lift gate **103** of the vehicle when the lift gate **103** is in a closed position in a case where the lift gate **103** is hinged at a top or side portion thereof. According to some aspects, when the lift gate **103** is opened, the opening **202** is uncovered by moving the lift gate **103** away from the opening **202**.

According to at least one aspect, the fascia door **201** includes a recess **204**. The recess **204** may function as a handle, where a user may push on a sidewall of the recess **204** to open or close the fascia door **201**. According to one or more aspects, the recess **204** includes a drain hole **205** in a lower portions, which may drain any liquid or debris that may enter the recess **204**.

FIG. **3** illustrates an embodiment of the vehicle including the rear fascia **100** including the fascia door **201** an open position and revealing a washer fluid reservoir **301** (e.g., a rear mounted washer fluid reservoir) and a filler door **302** for the washer fluid reservoir **301**. The fascia door **201** in the open position may reveal at least a portion of the opening **202** in the rear fascia **100**.

According to at least one aspect, the fascia door **201** may be disposed to close the opening in the horizontal portion **203** of the rear fascia **100** of the vehicle, concealing, for example, the washer fluid reservoir **301**. It should be understood that the horizontal portion **203** may be relatively horizontal as compared to other portions of the rear fascia **100**, and that the horizontal portion **203** need not be disposed perfectly horizontal.

According to some aspects, a fascia door assembly comprises a fascia door support bracket, a fascia door bracket, and the fascia door. According to one aspect and referring to FIG. **4**, the fascia door support bracket **401** attached to an interior surface **402** of the rear fascia **100**. According to one aspect and referring to FIG. **5**, the fascia door **201** may be engaged with the fascia door bracket **501**. According to some aspects and referring to FIG. **6**, the fascia door bracket **501** may be engaged with the fascia door support bracket **401**.

According to one aspect and referring to FIG. **4**, the fascia door support bracket **401** may be attached to an interior surface **402** of the rear fascia **100**, below the horizontal portion **203** of the rear fascia of the vehicle. The fascia door support bracket **401** may be attached to an interior surface **402** of the rear fascia **100** by, for example, an adhesive based bond, mechanical fastenings, ultrasonic assembly, metal inserts, snap fits, electromagnetic and heat welding, etc. The fascia door support bracket **401** may extend along the interior surface of the rear fascia to one or more generally vertical portions of the rear fascia, e.g., to an interior surface of the upper fascia **403**, toward a peripheral edge **404** of an opening in the rear fascia **100** extending upward from the horizontal portion. According to some aspects, the fascia door support bracket **401** may include a support bracket opening **405**.

According to at least one aspect, the support bracket opening **405** may be coincident with the opening **202** in the rear fascia **100**. For example, the fascia door support bracket **401** may be attached to an interior surface **402** of the rear fascia **100**, and the support bracket opening **405** may be disposed to allow access to the washer fluid reservoir through the opening **202**. The support bracket opening **405** may be larger, the same size, or smaller than the opening **202** in the rear fascia **100**.

According to one aspect and referring to FIG. **4**, the fascia door support bracket **401** may improve the rigidity of the rear fascia **100**. For example, the rear fascia **100** and the fascia door support bracket **401** may have a structural rigidity sufficient to withstand loads typically experienced when loading and unloading the vehicle, e.g., with luggage.

According to one aspect and referring to FIG. **5**, the fascia door bracket **501** includes a bracket opening **502** and rails (e.g., a first rail **503** and a second rail **504**) along at least two parallel edges of the bracket opening **502**.

According to one aspect, the fascia door **201** includes at least two projection portions that may be engaged with the rails of the fascia door bracket **501**. For example, the projection portions may be "C" channels, which mate with the rails, enabling the fascia door to slide along a length of the opening. According to some aspects, a rail fits between projection portions of a C channel.

While FIG. **5** illustrates an embodiment in which the fascia door **201** includes the C channels and the fascia door bracket **501** includes the rails, the C channels and the rails may be arranged with the fascia door bracket **501** and the fascia door **201**, respectively.

According to one aspect and referring again to FIG. **5**, the fascia door **201** and the fascia door bracket **501** may each include detents, e.g., fascia door detents **505** and fascia door bracket detects **506**, which may engage as the fascia door is moved along the opening, such that the fascia door may be positioned and held in place (e.g., in an open, intermediate, or closed position). In one aspect, the fascia door detents **505** are disposed on an inner surface of the C channel, and the C channel may be resilient, such that the fascia door **201** may be moved along the opening, overcoming an engagement of the detents by a deformation of the C channel.

According to some embodiments and referring to FIG. **6**, the bracket opening **502** includes a longitudinal portion **601** and a traverse portion **602**. The longitudinal portion **601** may extend along a direction of travel of the fascia door **201**. The traverse portion **602** may be an opening at an end of the longitudinal portion **601**. The traverse portion **602** may enable the C channels of the fascia door **201** to be engaged with the rails, which may extend along the longitudinal portion **601**. According to one aspect, first ends of the rails are accessible at the traverse portion **602**, such that the C channels of the fascia door **201** may be engaged with the rails.

According to some aspects and referring again to FIG. **7**, the fascia door bracket **501** may be coupled with the fascia door support bracket **401**, for example, by connecting members (e.g., first connecting member **701**). According to some aspects, the connecting members may be bolts, screws, snaps, push type scrivet retainers, etc. According to one or more aspects, the fascia door support bracket **401** includes connecting member receiving portions (see for example, FIG. **4**, first connecting member receiving portion **406**). The connecting member receiving portions may be engaged by the connecting members, securing the fascia door bracket **501** to the fascia door support bracket **401**.

According to one aspect and referring again to FIG. **7**, the fascia door bracket **501** fits over the fascia door support bracket **401**, and may include a first support bracket vertical portion **702**, a second support bracket vertical portion **703**, and a support bracket middle portion **704** between the first and second support bracket vertical portions. It should be understood that the support bracket vertical portions are relatively vertical as compared to other portions of the rear fascia **100** or the support bracket middle portion **704**, and that the vertical portions need not be disposed perfectly vertical.

According to at least one aspect and as illustrated in FIG. **7**, the bracket opening **502** (see FIG. **5**) may be coincident with the opening **202** in the rear fascia **100** and the support bracket opening **405**, enabling the fascia door **201** to be moved between an open position and a closed position. For example, the fascia door bracket **501** may be mounted to the fascia door support bracket **401**, and the fascia door support bracket **401** may be attached to an interior surface **402** of the rear fascia **100**, wherein the bracket opening **502** and the support bracket opening **405** may be disposed to allow access to the washer fluid reservoir through the opening **202** in the rear fascia when the fascia door **201** is in an open position. The bracket opening **502** may be larger, the same size, or smaller than the opening **202** in the rear fascia **100** and the support bracket opening **405**.

According to some aspects, the fascia door **201** includes a resilient catch **705** at an end portion. According to one aspect, the resilient catch **705** may engage with a detent **706** of the fascia door bracket **501** or the horizontal portion **203** of the rear fascia when the fascia door **201** is in an opened position. According to some aspects, the fascia door **201** may include a second resilient catch **303** at another end portion (see FIG. **3**) that may engage with a detent of the fascia door bracket or the horizontal portion of the rear fascia when the fascia door **201** is in a closed position.

According to some aspects and referring to FIG. **8**, the fascia door assembly comprises the fascia door support bracket **401**, the fascia door bracket **501**, and the fascia door **201**. According to one aspect, a washer fluid reservoir **801** is disposed below the fascia door **201**. The washer fluid reservoir **801** may include a filler neck portion **802** and a filler door (not shown).

According to one aspect and referring to FIG. **8**, the fascia door support bracket **401** may be attached to the interior surface of the rear fascia. According to one or more aspects, the fascia door support bracket **401** may extend along the interior surface of the rear fascia to the interior surface of the upper fascia (e.g., first interior surface **803**) and the interior surface extending upward from the horizontal portion (e.g., second interior surface **804**). According to one aspect, the fascia door bracket **501**, including the first support bracket vertical portion **702**, the second support bracket vertical portion **703**, and the support bracket middle portion **704** between the first and second support bracket vertical portions, fits over the fascia door support bracket **401**. According to some aspects, the fascia door bracket **501** may be secured to the fascia door support bracket **401** by connecting members and connect member receiving portions, for example, a second connecting member **807** and second connecting member receiving portion **808**.

According to some aspects and referring again to FIG. **8**, the fascia door **201** includes a first projection portion **805** and a second projection portion **806**, and the fascia door bracket **501** includes the first rail **503** and the second rail **504**. According to some aspects, the rails engage with respective projection portions (e.g., first rail **503** engages with first projection portion **805**).

According to different aspects, an upper surface **809** of the fascia door **201** may be have a larger area than the opening **202** (as illustrated in FIG. **8**), or may have an area that is about the same size, or smaller, than the opening **202**, wherein the upper surface **809** may fit inside of the opening **202** when the fascia door **201** is in the closed position.

FIG. **9** illustrates an embodiment of the washer fluid reservoir **301**, the filler door **302**, and the filler neck portion **802**. FIG. **9** further illustrates a user's finger **901** reaching through the opening in the rear fascia to manipulate (e.g., open or close) the filler door **302** (e.g., by releasing a catch). According to some aspects, the filler door **302** is arranged to open in a direction of the fascia door **201** in an open position in the opening of the rear fascia. According to some aspects, the fascia door may be arranged to close the filler door when the fascia door is moved from the open position to the closed position in the case where the filler door opens toward the fascia door in the open position so a closed state of the filler door **302** can be determined based on the closed position of the fascia door. In some aspects, the filler door **302** opens away from the direction of the fascia door **201** in an open position in the opening of the rear fascia.

According to some aspects, the fascia door is formed on a resilient material that conforms to a requirement for rear fascia, e.g., deflection under load, deflection in a steel ball drop test, etc. According to one aspect, the fascia door assembly including the fascia door support bracket, the fascia door bracket, and the fascia door are configured to withstand loads typically experienced when loading and unloading the vehicle, e.g., with luggage.

While various embodiments of the disclosure have been described, the description is intended to be exemplary, rather than limiting and it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible that are within the scope of the disclosure. Accordingly, the disclosure is not to be restricted except in light of the attached claims and their equivalents. Also, various modifications and changes may be made within the scope of the attached claims.

We claim:

1. A fascia door assembly comprising:
- a fascia door support bracket having a support bracket opening;
- a fascia door bracket coupled to the fascia door support bracket and having a bracket opening; and
- a fascia door engaged with the fascia door bracket in the bracket opening, the fascia door having an opened position in the bracket opening and a closed position in the bracket opening;
- wherein the opening of the rear fascia is in a horizontal portion of the rear fascia and access to the opening depends on a position of a lift gate of the vehicle.

2. The fascia door assembly of claim **1**, wherein the fascia door support bracket is attached to an interior surface of a rear fascia of a vehicle, and an opening in the rear fascia is coincident with a portion of the bracket opening.

3. The fascia door assembly of claim **2**, wherein the fascia door has an open position and a closed position, and wherein in the closed position the opening in the rear fascia is closed by the fascia door.

4. The fascia door assembly of claim **1**, wherein the fascia door comprises a recess.

5. The fascia door assembly of claim **1**, wherein the fascia door in a closed position is above a filler door of a washer fluid reservoir.

6. The fascia door assembly of claim **5**, wherein the washer fluid reservoir is below a rear fascia of a vehicle.

7. A fascia door assembly comprising:
- a fascia door support bracket having a support bracket opening;
- a fascia door bracket coupled to the fascia door support bracket and having a bracket opening; and
- a fascia door engaged with the fascia door bracket in the bracket opening, the fascia door having an opened position in the bracket opening and a closed position in the bracket opening;
- wherein the fascia door comprises a plurality of projections engaged with respective rails of the fascia door bracket.

8. The fascia door assembly of claim **7**, wherein the fascia door comprises a first detent and the fascia door bracket comprises a second detent engaged with the first detent in at least one of the opened position and the closed position.

9. The fascia door assembly of claim **1**, wherein the fascia door comprises a resilient catch at an end portion of the fascia door and the fascia door bracket comprises a detent engaged with the resilient catch in one of the closed position and the opened position.

10. A vehicle comprising:

a rear fascia having an opening in a horizontal portion of the rear fascia;

a washer fluid reservoir behind the rear fascia and below the opening; and

a fascia door, wherein the washer fluid reservoir is concealed below the fascia door in a closed position and revealed by movement of the fascia door to an open position.

11. The vehicle of claim **10**, further comprising:

a fascia door support bracket attached to an interior surface of the rear fascia and having a support bracket opening; and

a fascia door bracket coupled to the fascia door support bracket and having a bracket opening coincident with the opening in the rear fascia and the support bracket opening, wherein the fascia door is engaged with the fascia door bracket.

12. A vehicle comprising:

a rear fascia having an opening;

a washer fluid reservoir behind the rear fascia; and

a fascia door assembly comprising:

a fascia door support bracket attached to an interior surface of the rear fascia and having a support bracket opening;

a fascia door bracket coupled to the fascia door support bracket and having a bracket opening coincident with the opening in the rear fascia and the support bracket opening; and

a fascia door engaged with the fascia door bracket in the bracket opening, wherein the washer fluid reservoir is concealed below the fascia door in the opening in the rear fascia.

13. The vehicle of claim **12**, wherein the fascia door has an open position in the bracket opening and a closed position in the bracket opening closing the opening in the rear fascia.

14. The vehicle of claim **12**, wherein the opening of the rear fascia is in a horizontal portion of the rear fascia, wherein access to the opening is concealed by a lift gate of the vehicle in a closed position.

15. The vehicle of claim **12**, wherein the opening of the rear fascia is in a horizontal portion of the rear fascia, wherein access to the opening is enabled by a lift gate of the vehicle in an opened position.

16. The vehicle of claim **12**, wherein the fascia door comprises a recess.

17. The vehicle of claim **12**, wherein the fascia door in a closed position is above a filler door of a washer fluid reservoir.

18. The vehicle of claim **12**, wherein the washer fluid reservoir comprises a filler cap, the filler cap configured to open towards the fascia door in an opened position and configured to be closed by the fascia door when the fascia door is moved from the opened position to a closed position.

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